COURSE OUTLINE

Course Code: SE 444	**Course Title:** Introduction to Artificial Intelligence	**Semester:** Fall 2025
Course Credits: 3	**Lectures Hours: 3 Hrs.**	**Tutorial Hours:** 0
Lectures Schedule: **Location:**	**Male Section: MON/WED, 13:00 to 14:15, BG.069** **Female Section: MON/WED, 10:30 to 11:50, BG.068**	
Course Instructors: **Name:** **Office Location:** **Email:** **Office Phone:** **Office Hours:**	Prof. Anis Koubaa SG-10 akoubaa@alfaisal.edu **7497** **MON/WED** 12:00 PM – 01:00 PM or 02:30 PM to 03:30 PM (or appointment)	

Required Textbook:

Russell & Norvig, *Artificial Intelligence: A Modern Approach (4th Edition),* ***2020***

Notes and other supplemental materials:
http://elearning.alfaisal.edu/

Prerequisites:	**Co-requisites:** None	**Course Type:** ■ Required ☐ Elective ☐ Selected Elective

Course Description:

This course introduces students to the fundamental ideas, methods, and applications of Artificial Intelligence (AI). It explores how intelligent systems are designed to perceive, reason, and act in complex environments. Core topics include problem-solving, search techniques, knowledge representation, logical reasoning, planning, and decision-making under uncertainty.

The course also highlights concepts of multi-agent systems, game playing, and the role of AI in modern applications such as robotics, natural language processing, and decision-support tools. Emphasis is placed on developing problem-solving and analytical skills through practical exercises and programming assignments. Students will gain an understanding of how AI methods can be applied to real-world problems in engineering, business, and society. By the end of the course, students will be equipped with foundational knowledge of AI principles and the ability to critically assess the capabilities, limitations, and ethical considerations of intelligent systems.

Major Topics:

	Topic
1.	Introduction to AI & Agents
2.	Search (uninformed & informed)
3.	Games & Adversarial Search
4.	Constraint Satisfaction Problems
5.	Logic & Knowledge Representation
6.	Planning
7.	Probability & Reasoning under Uncertainty
8.	Bayesian Networks & Decision Theory
9.	Multi-Agent Systems & Game Theory
10.	AI Applications & Ethics

* Calendar and outline are tentative and subject to change depending on unforeseen influential factors.

#	CLO	Course Learning Outcomes (CLO) *Students who successfully complete this course should be able to:*	ABET Outcomes (i.e. addressed by the course)
1.0		**Knowledge & understanding**	
1.1	1	**Explain** core AI concepts, including search, knowledge representation, reasoning, planning, and uncertainty.	SO 1
2.0		**Skills**	
2.1	2	**Implement** AI-based solutions for problems such as search, games, constraint satisfaction, and probabilistic reasoning using programming tools.	SO 2, SO 6
3.0		**Values, autonomy, and responsibility**	
3.1	3	**Evaluate** the ethical, societal, and professional implications of AI in engineering and real-world applications.	SO 3, SO 4, SO 5
3.2	4	**Collaborate** effectively as a member or leader of a team to design and deliver AI-based solutions**.**	

Schedule

Class Meetings: Mondays & Wednesdays, 1.5 hrs each
Start: Mon 25 Aug 2025
End: Wed 17 Dec 2025

Week	Date Range	Day	Topic	Assessment / Activity
1	Aug 25–27	Mon	Introduction to AI, intelligent agents	—
		Wed	Agents & environments, rationality	—
2	Sept 1–3	Mon	Problem formulation, BFS, DFS	Assignment 1 released
		Wed	UCS, practice problems	Lab (Search coding)
3	Sept 8–10	Mon	Informed search (Greedy, A*)	—
		Wed	Heuristics, A* lab	**Quiz 1 (Search)**
4	Sept 15–17	Mon	Local and Adversarial search (Minimax)	—
		Wed	Tutorial on Local and Adversarial search	Lab
5	Sept 22–24	Mon	Constraint Satisfaction Problems (CSP)	Assignment 2 released
		Wed	Backtracking, Sudoku/map coloring	Lab
6	Sept 29–Oct 1	Mon	Review & practice (Search + CSP)	—
		Wed	**Midterm 1** (Weeks 1–5)	Exam
7	Oct 6–8	Mon	Propositional logic, inference	—
		Wed	Knowledge representation basics	Lab
8	Oct 13–15	Mon	First-order logic, unification	Assignment 3 released
		Wed	Logic lab (KB examples, medical diagnosis)	Lab
9	Oct 20–22	Mon	Planning (STRIPS, robot navigation)	—
		Wed	Planning lab, problem solving	**Quiz 2 (Logic + Planning)**
10	Oct 27–29	Mon	Probability basics, Bayes rule	Assignment 4 released
		Wed	Naïve Bayes, spam filter demo	Lab
11	Nov 3–5	Mon	Review & practice (Logic + Probability)	—
		Wed	**Midterm 2** (Weeks 7–10)	Exam
12	Nov 10–12	Mon	Bayesian networks, inference	—
		Wed	Decision theory, applications	Lab
13	Nov 17–19	Mon	Multi-agent systems, game theory	Assignment 5 released
		Wed	Prisoner's dilemma, auctions	Lab

14	Nov 24–30	**Sun–Sat**	**Mid-Semester Break (No Classes)**	—
15	Dec 1–3	Mon	AI applications (Robotics, NLP, Ethics, Vision 2030)	—
	Dec 8–10	Mon	Project development + mentoring	—
	Dec 15–17	Wed	**Final Project Presentations & Wrap-up**	Project due

Course Assessment Methods & Weights

Assessment Component	Weight	Details
Midterm Exam 1	20%	Covers Weeks 1–5 (Search, Games, CSP).
Midterm Exam 2	20%	Covers Weeks 7–10 (Logic, Planning, Probability).
Final Exam	30%	Comprehensive.
Quizzes (2)	20%	Quiz 1 (Search), Quiz 2 (Logic & Planning).
Project (2)	10%	Short programming/problem-solving tasks (Search, CSP, Logic, Bayes, Game Theory), bayesian decision making.

Alfaisal University Grading Scale

Grade	Percent Range	Grade	Percent Range
A+	95%-100%	C+	75%-79%
A	90%-94%	C	70%-74%
B+	85%-89%	D+	65%-69%
B	80%-84%	D	60%-64%
		F	0%-59%

Program Learning Outcomes / Student Outcomes

K1 PLO1 SO8	an ability to recognize the integrated body of knowledge in engineering & understand the underlying theories, technologies, principles, concepts, processes, materials, techniques, practices, conventions and/or terminology in engineering; as well as engineering research methodology and inquiry techniques.
S1 PLO2 SO1	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
S2 PLO3 SO7	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
S3 PLO4 SO2	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
S4 PLO5 SO6	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
S5 PLO6 SO3	an ability to communicate effectively with a range of audiences
V1 PLO7 SO4	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
V2 PLO8 SO5	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

Policies and Notes

1) Student Affairs Notes

- The course follows the University policy as outlined in the Undergraduate Student Handbook.
- Sessional dates, registration deadlines, and withdrawal deadlines are available on the **Student Affairs and Admission** website.
- The **final exam date** will be announced by Student Affairs towards the end of the semester.
- Once deadlines and exam dates are published, no rescheduling will be permitted, except in cases of unforeseeable circumstances where the university suspends classes. In such cases, in-session submissions and exams will be automatically rescheduled to the following session.

2) Attendance Policy

- **Class Attendance:** Students are expected to attend all lectures, labs, and tutorials. Exceeding **20% of lecture-hour absences** may result in a **Dismissal Notification (DN) grade**. The 20% allowance covers medical emergencies or the death of an immediate family member, with valid supporting documentation required. Only in these cases can students make up missed coursework or exams. Excessive absenteeism may result in loss of scholarship eligibility or administrative withdrawal from the course and University.
- **Late Arrival:** Arriving **5 minutes late** counts as tardy, with **three tardies equaling one absence**. Arriving **10 minutes late or more** is recorded as a full absence.

3) Electronic Devices Policy

- Use of mobile phones, laptops, tablets, or other portable electronic devices during class is **prohibited** unless explicitly permitted by the instructor. Violation may result in being asked to leave class and receiving a full absence.
- Electronic Devices are **prohibited during exams**, except calculators approved by the instructor. Mobile phones, even if switched off, will result in a score of **zero (0)** and may incur further disciplinary action.

4) Late Submission of Student Work

- Homework, assignments, lab reports, and projects must be submitted **on time**. Late submissions will be penalized according to the instructor's policy and **will not be accepted more than two days past the deadline**, including weekends.

5) Academic Integrity

- Cheating, plagiarism, or any form of academic dishonesty constitutes a **scholastic offense**. Cheating in examination or attempting to cheat, or violating instructions, and examination regulations shall render the student to punishment in accordance with the Student Disciplinary By-Laws as issued by the University Council. Penalties may include expulsion from the program. Students are expected to maintain academic integrity and consult the instructor if uncertain.

6) E-Learning Management System

- **Moodle** is the official platform for course materials, announcements, and messages. Students are responsible for regularly checking Moodle for updates.

7) Instructional Materials

- Instructors are expected to post all instructional materials no later than two days after the lecture.

8) Lines of Communication

- Students should follow the established communication chain and raise issues in the following order:
  1. **Course Instructor/Professor**
  2. **Department Chair** (if unresolved in reasonable time)
  3. **Vice Dean for Academic & Student Affairs** (if the issue merits escalation)
  4. **Dean** (issues may then be addressed by the Dean or escalated to the President at the Dean's discretion)

9) Exams and Retake Policy

1. Students are not allowed to enter the examination venue if more than 5 minutes have passed since the start of the exam.
2. Students are required to remain in the examination venue for a minimum of 30 minutes before they are permitted to leave.
3. Students must possess a valid student ID in order to be allowed entry into any exams.
4. In case a student misses a major exam, the student must **submit an Absence Leave eForm** with supporting documentation **no later than one week** after missing the exam.
5. If a student has an approved absence (via e-form), the faculty member must provide **a retake exam**—covering the same material as the original—**within two weeks**.
6. Alternatives such as transferring the exam weight to another assessment or substituting it with another grade component **are not permitted**.

10) Accessibility and Special Accommodations

- In accordance with the University's Undergraduate Student Handbook, students with documented disabilities or special needs are entitled to reasonable accommodations. Students requiring accommodations, such as adjustments in coursework or exams, must contact the Deanship of Student Affairs and follow the University's Examination Special Accommodation Policy (Section 3.32). Students are advised to notify the instructor as early as possible in the semester to ensure timely arrangements. Counseling and support services are also available through the University's Counseling & Skills Development unit (Section 5).